PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: ScrapSync: AI-Powered Industrial Symbiosis & Decision Intelligence Platform

Version: 1.0

1. Project Overview

- **Problem Statement:**

  Industrial manufacturers, especially SMEs, generate large volumes of byproduct waste such as wood dust, metal scraps, and offcuts. These materials are often discarded due to the lack of visibility into their potential economic value and absence of efficient matching systems with buyers.

  Waste data is typically stored in unstructured formats such as handwritten logs, receipts, or images, making it difficult to analyse and utilise. As a result, businesses rely on manual, intuition-based decisions, leading to unnecessary disposal costs and missed revenue opportunities.

- **Target Domain:** Domain 2: AI for Economic Empowerment & Decision Intelligence

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- **Proposed Solution Summary:**

ScrapSync is an AI-powered decision intelligence platform that helps manufacturers transform industrial waste into a profitable resource. Instead of manually handling waste data and struggling to find buyers, the system automates analysis and provides smart recommendations, allowing companies to make better, data-driven decisions.

The platform integrates marketplace connectivity with advanced AI technologies to simplify the entire waste management and selling process. By leveraging Google Gemini 2.5, ScrapSync not only processes data but also interprets it to deliver meaningful insights for business use.

Key capabilities of ScrapSync include:

- **Connecting manufacturers with waste buyers**
  ScrapSync acts as a digital marketplace that links manufacturers to verified buyers who are interested in specific types of waste materials, reducing the time and effort needed to search for partners.

- **AI-powered waste data analysis (OCR + AI)**
  The system uses Optical Character Recognition (OCR) to extract data from documents such as receipts, invoices, or logs, and then applies AI to analyze waste type, quantity, and patterns automatically.

- **Generating actionable insights and forecasts**
  ScrapSync processes collected data to provide insights such as waste value estimation, demand trends, and future price predictions, helping users understand the true potential of their waste.

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- **Smart decision support for selling waste**

  The platform recommends:

  - the best buyers based on demand, pricing, and location
  - the optimal time to sell based on market trends
  - the most profitable selling strategy

- **Maximizing waste value and profitability**

  By combining all these features, ScrapSync enables manufacturers to:

  - accurately determine the value of their waste
  - identify high-value opportunities
  - make informed decisions to maximize profit and reduce losses

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2. Background & Business Objective

- **Background of the Problem:**

Industrial waste management in SMEs and factories is often handled through manual and intuition-based decisions.

Most companies:

- treat waste as a liability rather than an asset
- lack access to real-time pricing or demand data
- rely on fixed disposal processes (e.g., dumpsters, contractors)

Furthermore:

- waste data exists in fragmented formats (images, receipts, logs)
- companies do not have tools to interpret and convert this data into insights
- As a result, businesses lose potential revenue and incur unnecessary cost

- **Importance of Solving This Issue:**

Solving this problem can:

- transform waste into a new revenue stream
- reduce operational costs for SMEs
- improve efficiency in resource utilisation
- support sustainable practices by reducing landfill waste

From an economic perspective:

- SMEs can increase profitability
- buyers gain access to cheaper raw materials
- overall market efficiency improves

This directly aligns with the need for data-driven decision-making in modern industries.

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- **Strategic Fit / Impact:**

  ScrapSync aligns strongly with the AI for Economic Empowerment & Decision Intelligence domain by:

  - Enabling data-driven decisions instead of manual judgement
  - Using Google Gemini 2.5 to:
    - interpret structured and unstructured data (OCR + inputs)
    - generate insights (value estimation, trends)
    - recommend actions (best buyer, pricing strategy)
    - explain decisions clearly to users

  Expected Impact:

  - Reduction in disposal costs (e.g., RM500 → RM0/month)
  - Generation of new revenue streams from waste
  - Faster and smarter decision-making
  - Increased economic efficiency in industrial ecosystems

  If the AI component is removed:

  - the system becomes a simple listing platform
  - no intelligent insights or decision support will be available

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3. Product Purpose

- **3.1. Main Goal of the System:**
  To provide manufacturers and industrial SMEs with an AI-powered platform that transforms unstructured waste data into profitable transactions — automating waste classification, buyer matching, logistics planning, and ESG reporting in a single unified system.

- **3.2. Intended Users (Target Audience):**
  - Manufacturer / Factory Manager
  - Buyer / Recycler
  - Admin / Platform Operator
  - Warehouse / Operations Manager
  - Sustainability / ESG Reporting Officer
  - Supply Chain / Logistics Coordinator

4. System Functionalities

- **4.1. Description:**
  ScrapSync operates as an AI-powered industrial symbiosis platform built on a Streamlit frontend, FastAPI backend, and Supabase (PostgreSQL) database. At its core, Google Gemini 2.5 Flash serves as a multimodal AI engine that reads unstructured waste documents (photos, handwritten logs, receipts) and converts them into structured JSON data in a single step — bypassing brittle traditional OCR entirely. The system then cross-references this data against a live buyer network, generates profitable logistics routes, and produces smart contracts and ESG reports automatically.

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- **4.2. Key Functionalities:**
  - **AI Synergy Engine (Instant Matchmaking):** A user uploads a photo of a waste log; Gemini reads it multimodally, classifies the material (e.g., Wood Dust), and scans the live Supabase buyer database to identify the highest-paying buyer. It then calculates projected profit, maps the logistics route, drafts a Digital Twin Smart Contract, and records the signed deal directly into the live ledger.
  - **Enterprise Batch Processing:** Warehouse managers can drag and drop up to 50 waste receipts at once. The system processes all uploads in the background, aggregates the data, and outputs a consolidated "Monthly Projected Profit" dashboard, demonstrating the platform's enterprise-grade scalability.
  - **Predictive AI Insights & Forecasting:** ScrapSync proactively analyses historical ledger data to detect waste generation patterns and market price trends. It alerts managers with actionable recommendations such as auto-booking logistics early to lock in higher prices before market rates drop, transforming the platform from a passive tool into a revenue-generating asset.
  - **ESG Dashboard & Corporate Ledger:** Every signed smart contract is recorded into an immutable Supabase ledger. The system automatically translates transaction data into live ESG metrics including Carbon Avoided and Landfill Diverted, enabling factories to automate their government sustainability reporting obligations without manual data entry.

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4.3. AI Model & Prompt Design

4.3.1. Model Selection: Gemini 2.5 Flash

ScrapSync has transitioned to **Gemini 2.5 Flash** as its primary AI engine. This model was chosen specifically to move beyond the limitations of traditional, brittle OCR systems (e.g., Tesseract). Key advantages include:

- **Native Multimodal Reasoning:** Gemini 2.5 Flash does not just "read" text; it "perceives" the document. It can interpret messy handwriting, understand the context of non-standard industrial logs, and handle low-light receipt photos in a single pass.
- **Single-Step Inference:** Unlike multi-layered pipelines that require separate OCR, cleaning, and extraction steps, Gemini 2.5 Flash performs cross-referencing against our database and data extraction simultaneously.
- **Efficiency at Scale:** The "Flash" variant provides the high-speed processing necessary for the platform's batch-upload feature (up to 50 files) without the latency typically associated with large-scale multimodal models.

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4.3.2. Prompting Strategy

ScrapSync utilizes a multi-step agentic prompting strategy to ensure the "AI Brain" remains focused and verifiable:

1. **Direct JSON Extraction (Step 1):** The raw image is passed to Gemini with a system instruction to extract waste type, quantity, unit, and condition. By leveraging Gemini's native ability to output structured data, the system bypasses the need for post-processing regex or formatting scripts.
2. **Contextual Matching (Step 2):** The extracted JSON is injected into a second prompt containing a live snapshot of the Supabase buyer database. Gemini cross-references the waste grade against buyer demand profiles to generate a compatibility ranking and matching rationale.
3. **Autonomous Contract Drafting (Step 3):** The final match triggers a third prompt where Gemini generates a Digital Twin Smart Contract, pre-filling complex transaction terms based on the confirmed data.

4.3.3. Context & Input Handling

ScrapSync processes multimodal inputs including JPEG, PNG, and PDF formats.

- **Zero Pre-processing:** Images are base64-encoded and passed directly to the Gemini 2.5 Flash multimodal endpoint. By avoiding traditional OCR pre-processing, we eliminate a common point of failure and preserve the "raw" visual context of the logs.
- **Robust Batch Processing:** The system handles up to 50 files (max 10MB each) per batch. If the model cannot parse a specific file (e.g., severe corruption), that instance is flagged in the report while the rest of the batch completes, ensuring a seamless user experience.

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4.3.4. Fallback & Failure Behavior

To ensure 100% workflow uptime, ScrapSync implements the following fail-safes:

- **Constraint Reinforcement:** If a response fails JSON validation, the system triggers an automatic retry with a reinforced prompt that constrains the output space.
- **Manual Override:** If Gemini cannot confidently parse a document after two attempts, the system falls back to a manual entry form, ensuring the industrial process is never stalled by an AI hallucination.
- **Database Integrity:** Every Buyer ID generated in Step 2 is validated against the Supabase demand_nodes table. If Gemini suggests a non-existent buyer, the match is discarded and the next valid ranking is presented.

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5. User Stories & Use Cases

- **User Stories:**
  - "As a factory manager, I want to upload a photo of my waste log so that the system automatically identifies the material and finds the best buyer for me."
  - "As a warehouse manager, I want to upload 50 waste receipts at once so that I can see my total monthly projected profit without manual calculation."
  - "As a manufacturer, I want to receive proactive alerts about dropping market prices so that I can auto-book logistics and lock in today's higher rate."
  - "As a compliance officer, I want the system to automatically generate ESG reports from transaction data so that I can meet government sustainability reporting requirements without extra effort."
- **Use Cases (Main Interactions):**

  Waste Upload & Synergy Flow:
  Manufacturers upload a waste log image via the Streamlit dashboard. FastAPI routes the file to Google Gemini 2.5, which extracts waste type, quantity, and condition as JSON. The system queries Supabase demand_nodes for the best-matching buyer, displays the match with AI reasoning and projected profit (e.g., RM 24,000), and prompts the user to sign a Digital Twin Smart Contract. Upon signing, the transaction is written to the Supabase ledger and ESG metrics are updated in real time.

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Batch Processing Flow:

Warehouse manager drags and drops a folder of up to 50 waste receipts. The system queues all files and processes each asynchronously through the same AI extraction pipeline. Once complete, the dashboard aggregates all results and displays a consolidated Monthly Projected Profit summary, a breakdown by waste type, and flags any files that failed processing.

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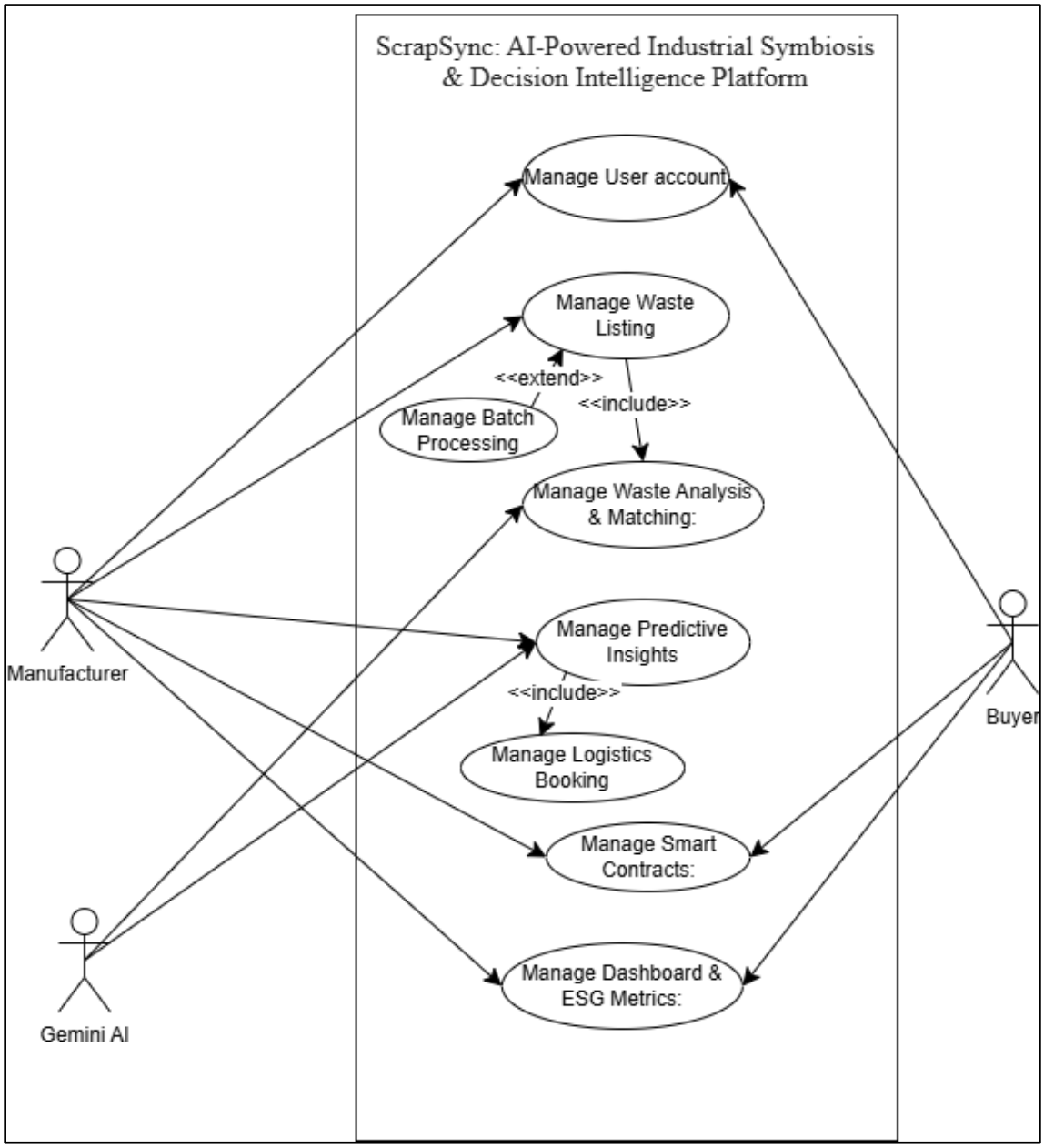
- **Use Case Diagram:**



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6. Features Included (Scope Definition)

- Multimodal AI waste data extraction from images, handwritten logs, receipts, and scanned documents using Google Gemini 2.5 (Module 2 & 3)
- AI-powered smart buyer matching with ranked recommendations, match reasoning, and projected profit calculation (Module 4)
- Enterprise batch processing of up to 50 waste documents simultaneously with aggregated monthly profit dashboard (Module 2)
- Digital Twin Smart Contract generation and immutable Supabase transaction ledger with live ESG metrics (Carbon Avoided, Landfill Diverted) (Module 6 & 7)

7. Features Not Included (Scope Control)

- User registration and role selection flow (account creation page not yet implemented; login session exists but full registration is out of scope for this version)
- Full transaction negotiation workflow (offer, counter-offer, accept/reject cycle between manufacturer and buyer is not yet implemented; only the contract signing button UI exists)
- Admin management dashboard (company verification, listing moderation, and platform-wide monitoring are not implemented in this version; basic diagnostics only)

8. Assumptions & Constraints

- **LLM Cost Constraint:** Each average user session (single waste upload + buyer match + contract generation) consumes approximately 2,000–4,000 tokens across three chained prompts. For batch sessions of 50 documents, estimated token usage is 100,000–200,000 tokens per session. To manage inference costs at scale, ScrapSync caches the buyer demand_nodes snapshot from Supabase at the start of each session and reuses it across all files in a batch, rather than re-querying the database and re-injecting the full buyer list into every individual prompt. This reduces redundant token usage significantly for high-volume batch jobs.

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- **Technical Constraints:** The platform currently accepts only image files (JPEG, PNG) and PDF documents as waste input. Direct text entry via a manual form is not yet supported. The system runs on Supabase's free tier, which limits concurrent database connections and row-level throughput; this may cause latency for large simultaneous user sessions. The Streamlit frontend does not yet support multi-user concurrent editing of the same listing.
- **Performance Constraints:** Each AI inference call (image extraction + buyer matching) takes approximately 3–8 seconds depending on image complexity and network latency. For batch jobs of 50 documents, total processing time may reach 5–10 minutes. Users are shown a live progress indicator during batch processing. Single-document uploads are expected to complete within 10 seconds under normal load. The system does not currently support real-time streaming of AI responses; results are displayed only after each full inference completes.
- **User Input:** All waste data extraction is triggered by a user uploading a file; the system does not autonomously pull or scan data without user action. Buyer matching and contract generation also require explicit user confirmation at each step before proceeding. The platform assumes that uploaded images are legible and contain valid waste information; severely blurred, corrupted, or irrelevant images will result in extraction failure and prompt the user to re-upload.

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9. Risks & Questions Throughout Development

- **Extraction Accuracy on Low-Quality Images:** How do we ensure the AI correctly reads handwritten or low-resolution waste logs? Poor image quality, inconsistent handwriting, or non-standard receipt formats may cause the model to extract incorrect material types or quantities, leading to inaccurate profit projections and wrong buyer matches. Mitigation: the system validates extracted JSON fields and flags low-confidence outputs for user review before proceeding to matching.
- **Buyer Hallucination & Stale Demand Data:** What happens if the AI recommends a buyer that no longer exists or has changed their accepted materials? The model may hallucinate buyer names or recommend outdated matches if the demand_nodes table is not kept current. Mitigation: every recommended buyer ID is validated against the live Supabase demand_nodes table before display; hallucinated or missing IDs are silently discarded and the next valid match is shown. Buyer records must be refreshed regularly by platform admins to maintain data accuracy.
- **ESG Metric Accuracy & Auditability**: How do we ensure that the Carbon Avoided and Landfill Diverted figures reported in the ESG dashboard are accurate enough for official government sustainability submissions? Currently, conversion factors (kg of waste to kg of CO2 avoided) are estimated using industry-standard coefficients, but these may not match the specific regulatory formulas required by Malaysian environmental authorities. Risk: companies relying on ScrapSync's ESG figures for compliance reporting may face audit challenges. Mitigation: clearly label all ESG figures as estimates and recommend independent verification before official submission; plan to integrate region-specific regulatory conversion tables in a future version.

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